STAFF REVIEW · H446 129 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Long consecutive runs of identical symbols.

2. 24.

3. RAM.

4. Sum 0, carry-out 1.

I Do — source worked steps

1. Two dictionary entries — Predict the first step from the prompt.

2. #1 #2 #1 — Explain the intermediate result or decision.

3. Restore original phrase — Treat spaces as retained separators: #1 #2 #1. Decoder substitutes entries to recover 'COMPUTER SCIENCE COMPUTER'. Include dictionary storage when calculating total size.

We Do — guided CHECK

1. ABCD repeats three times
Predict the first step from the prompt.
Token A for ABCD
Explain the intermediate result or decision.
Count dictionary overhead
AAXYZA, using single-character token A and leaving XYZ as literals under this toy scheme. Original 15 characters become 6 token/literal characters, plus dictionary overhead; real formats require unambiguous token conventions.

You Do — source mark schemes

1. Repeated substrings replaced by shorter references [3]; mapping resolves tokens for exact restoration [3].

2. 1 2 1 1 [3]; BLUE RED [3], with token separators assumed.

3. Repeated sequences help [2], tokens shorten recurring data [2], dictionary overhead matters [2], little redundancy in high-entropy data and justified choice [2].

Exit Ticket — source answer

1. False. The dictionary allows exact replacement when decoding.